

SYNTHETIC DEMO DOCUMENT · NOT FOR FIELD USE

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HPS-2500X Solar Inverter

Installation, Operations & Maintenance Manual

DOCUMENT	HPS-OM-2500X
VERSION	4.2
EFFECTIVE	March 1, 2026
STATUS	Approved — Synthetic
MANUFACTURER	Helios Power Systems (Synthetic)

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7. Thermal Management

The HPS-2500X controller continuously evaluates internal controller temperature, cooling-fan feedback, ambient conditions, and commanded power.

7.1 Protective operating states

STATE	CONTROLLER BEHAVIOR	OPERATOR MEANING
Full output	Normal power command is available.	No thermal restriction is active.
Derated	Power is reduced while the inverter remains online.	Protective response; not equivalent to a full trip.
Thermal trip	Power conversion stops and a manual review is required.	Higher-severity protection state.

7.2 Cooling-path inspection

- Confirm the work is authorized under the applicable site procedure.
- Check intake screens and exhaust paths for dust, vegetation, packaging, or standing water.
- Compare commanded and reported cooling-fan speed; investigate sustained variance.
- Do not bypass thermal limits or force an output command while a protective state is active.

Safety note: This synthetic manual does not replace site electrical-safety, isolation, or lockout/tagout requirements.

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7.3 Fault B17 — Controller Thermal Derating

Fault B17 is asserted when controller temperature exceeds the configured 75 °C derating threshold. The inverter remains connected and reduces active-power output to protect controller electronics.

Expected behavior: A B17 event changes the operating state from Full Output to Derated. It is a protective derating event, not evidence by itself of a failed controller or a full inverter trip.

Required checks before reset or return to full output

- 1 Verify controller temperature is trending downward and has remained below 70 °C for at least five minutes.
- 2 Inspect cooling-air intake and exhaust paths for obstruction or contamination.
- 3 Confirm cooling-fan feedback is present and does not show sustained variance from command.
- 4 Confirm the ambient-temperature input is plausible for site conditions.
- 5 Use the approved site procedure to acknowledge the event and restore service. Escalate repeated B17 events to reliability engineering.

Escalation criteria

- Controller temperature continues to rise after output is derated.
- Controller temperature reaches 82 °C.
- Cooling-fan feedback is absent or unstable.
- B17 repeats within 24 hours after an authorized inspection and reset.

Evidence interpretation: High ambient temperature can reduce thermal margin, but it does not by itself establish root cause. Inspection evidence is required before attributing the event to contamination or component failure.